outokumpu
high performance stainless steel

INSPECTION CERTIFICATE
EN 10204-3.1

Outokumpu Sales Order No.
9000123731

Customer Order No.
229185-TEM

Certificate No.
271620

Revision
00

Date
30 Oct 2024

Delivery No.
8000278710

Production mill
Avesta Plant

Sold to
Rolled Alloys, Inc.
125 West Sterns Road
TEMPERANCE MI 48182
USA

Product
Stainless Steel Hot Rolled Strip, Slit edge, Forta DX 2205

Finish
1D

Grade
1.4462
UNS S31803
UNS S32205

Ship to
Rolled Alloys
6630 Monclova Road
MAUMEE OH 43537
USA

Requirements
ASME SA-240/SA-240M BPVC.II.A-2023
ASTM A240/A240M - 23a
EN 10028-7:2016
NACE MR0103/ISO 17945:2015
NACE MR0175/EN ISO 15156-1:2020 & NACE MR0175/EN ISO 15156-3:2020

ASME SA-480/SA-480M BPVC.II.A-2023
ASTM A480/A480M - 23b
EN ISO 9444-2:2010

Mark of manufacturer: outokumpu

Process: E+AOD

Inspector's Stamp: AJA

Sales order item	Dimensions	Quantity (pcs)	Weight	Heat No.	Lot No.	Identification No. of Product	Customer's Material No.
20	0.2500 X 60.00 inch	3	47818 lbs	543487	1	543487-1	0010180

Chemical Composition

Heat No.	Country of Melt	C %	Si %	Mn %	P %	S %	Cr %	Ni %	Mo %	N %	PRE 16N
543487	Sweden	0.014	0.42	1.35	0.026	0.001	22.3	5.7	3.1	0.18	36

Test Results (1=Beginning, 2=Middle, 3=End, T=Transverse, L=Longitudinal, D=Diagonal, RT=Room temperature)

		EN ISO 6892-1 / ASTM A370 (RT)				ASTM E10 / EN ISO 6506-1	EN ISO 148-1
Identification No. of Product	Sample ID	RP0.2 ksi	RM ksi	A5 %	El 2" %	HARDNESS HBW	IMPACT VALUE 1 ft·lbf 20°C /68 °F
	Minimum	67	102	25	25		
	Maximum		138			293	
543487-1	1 T 853011	94	122	32	32	267	53
543487-1	3 T 853012	93	120	33	33	265	54

		EN ISO 148-1			ASTM A923-B		
Identification No. of Product	Sample ID	IMPACT VALUE 2 ft·lbf 20°C /68 °F	IMPACT VALUE 3 ft·lbf 20°C /68 °F	IMPACT VALUE MEAN ft·lbf 20°C /68 °F	IMPACT VALUE 1 ft·lbf -40°C /-40 °F	IMPACT VALUE 2 ft·lbf -40°C /-40 °F	IMPACT VALUE 3 ft·lbf -40°C /-40 °F
	Minimum			36	20	20	20
	Maximum						
543487-1	1 T 853011	56	59	58	45	47	42
543487-1	3 T 853012	59	56	58	44	44	44

		ASTM A923-B					ASTM A923-C
Identification No. of Product	Sample ID	IMPACT VALUE MEAN ft·lbf -40°C /-40 °F	LATERAL EXP. 1 in -40°C /-40 °F	LATERAL EXP. 2 in -40°C /-40 °F	LATERAL EXP. 3 in -40°C /-40 °F	LATERAL EXP. MEAN in -40°C /-40 °F	FERRIC CHLORIDE CORR. 24H 25°C /77 °F
	Minimum	20	0.015	0.015	0.015	0.015	
	Maximum						
543487-1	1 T 853011	44	0.047	0.048	0.038	0.044	OK

Outokumpu Stainless AB
Bergsnäsgatan 11
77441 Avesta, Sweden
Tel: +46 (0)226 810 00
Regoffice: Stockholm SWEDEN,
Regno: 556001-8748
VAT ID: SE556001874801

This material is found to comply with order requirements

Kåke Johansson
Kåke Johansson
Authorized Inspector
certificates.avesta@outokumpu.com

ACCEPTED
BY JC DATE 2/17/26

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Job: 089-25 PO: 40411 Line: 001 HT: 543487

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		ASTM A923-B					ASTM A923-C
Identification No. of Product	Sample ID	IMPACT VALUE MEAN ft·lbf -40°C /-40 °F	LATERAL EXP. 1 in -40°C /-40 °F	LATERAL EXP. 2 in -40°C /-40 °F	LATERAL EXP. 3 in -40°C /-40 °F	LATERAL EXP. MEAN in -40°C /-40 °F	FERRIC CHLORIDE CORR. 24H 25°C /77 °F
	Minimum	20	0.015	0.015	0.015	0.015	
	Maximum						
543487-1	3 T 853012	44	0.037	0.042	0.046	0.042	

		ASTM A923-C	FERRITESCOPE	PMI			
Identification No. of Product	Sample ID	CORR. RATE 24H mdd 25°C /77 °F	FERRITE CONTENT %	PMI			
	Minimum		35.0				
	Maximum	10	65.0				
543487-1	1 T 853011	0	48.0	OK			
543487-1	3 T 853012			OK			

Country of Origin: Sweden

Avg.t CO2e/t: 1.78

ISO 14067 validated, Cradle-to-gate

Solution annealed at 1070 °C/1958 °F. Rapidly cooled with water and air.

Certified acc. to PED 2014/68/EU by TÜV NORD, notified body, Reg-No. 0045

Charpy impact test (V-notch) subsize specimen 10x5 mm

Microstructure acc. To ASTM A923 method A: Unaffected (1 / T / 853011)

Marking, visual and dimensional inspection: Approved

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Packing Information

Delivery Item	Identification No. of Product	Batch No.	Quantity(pcs)	Weight	Length
10	543487-1	D051483030	1	15851 lbs	TM-1
10	543487-1	D051483130	1	15851 lbs	TM-2
10	543487-1	D051483230	1	16116 lbs	TM-3

1-HEAT # 543487-1
TRACER # 0718849US

2-HEAT # 543487-1
TRACER # 0718850US

3-HEAT # 543487-1
TRACER # 0718851US

ROLLED ALLOYS
Receiving Inspection
By:
Date: 12-4-24

Outokumpu Stainless AB Bergsnäsgatan 11 77441 Avesta, Sweden Tel: +46 (0)226 810 00 Regoffice: Stockholm SWEDEN, Regno: 556001-8748 VAT ID: SE556001874801	This material is found to comply with order requirements Kåke Johansson Kåke Johansson Authorized Inspector certificates.avesta@outokumpu.com	

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